

1. Why loops are necessary in Programming Language?

Loops are necessary because they allow us to **repeat a block of code multiple times** without writing the same code again and again.

For example, printing numbers from 1 to 10 can be done easily using a loop.

2. What is the purpose of pass keyword?

The pass keyword is used when we **do not want to write any code for a particular block yet**. It does nothing and simply allows the program to run without an error.

Example:

```
if age > 18:
```

```
    pass
```

3. What is the difference between break and continue?

break is used to **stop the loop completely**.

continue is used to **skip the current iteration** and move to the next iteration.

Example:

```
break    # Stops the loop
```

```
continue # Skips the current iteration
```

4. What is the difference between while loop and for loop?

A for loop is generally used when we **know how many times we want to repeat** something or when we want to go through items in a sequence.

A while loop is used when we want to **repeat the code as long as a condition is true**.

Example:

for loop

```
for i in range(5):
```

```
    print(i)
```

while loop

```
i = 0
```

```
while i < 5:
```

```
    print(i)
```

```
    i += 1
```